

Hampus Serneke

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EDUCATION

École polytechnique fédérale de Lausanne, EPFL

M.Sc. in Data Science

Lausanne, Switzerland

Sept. 2026 – July 2028

- **Current coursework:** machine learning, measure theory, math of data, foundations of data science, law of AI

Lund University

B.Sc. in Technology as part of the Engineering Mathematics program

Lund, Sweden

Aug. 2022 – June 2026

- **Cumulative GPA:** 4.9/5.0
- Pursued a major in Computer Science and Engineering for two years before transferring to Engineering Mathematics.
- **Thesis title:** Deep Learning for Rotation Averaging in Structure from Motion (GNNs in computer vision)
- **Coursework:** Mathematical statistics, matrix theory, stationary stochastic processes, applied mathematics (PDEs), algorithms and data structures, functional programming, object-oriented modelling, electromagnetic field theory

Katedralskolan Lund

Natural Science High School program

Lund, Sweden

Aug. 2018 – June 2021

- **GPA:** 22.5/22.5

WORK & LEADERSHIP EXPERIENCE

California Institute of Technology, Caltech

Summer Undergraduate Research Fellow

Pasadena, USA

June 2026 – Aug. 2026

- Currently co-authoring a paper with our results. Pre-print will be on arXiv soon
- Conducted research in reinforcement learning and AI safety at the Advanced Mechanical Bipedal Experimental Robotics Lab under the supervision of Dr. Pol Mestres and Prof. Aaron Ames
- Created the first hardware implementation of an anytime safe reinforcement learning algorithm (RL-SGF), going from theorem to hardware
- Ran hardware experiments on G1 humanoids and Crazyflie quadrotors using policies trained in IsaacLab and mjlab
- Implemented a novel learning rate scheduler for fulfilling the safety requirements, greatly increasing mean reward
- Obtained a sharper theoretical bound for the required learning rate to ensure safety
- Investigated the effects of relaxing the theory by introducing PPO-inspired estimates

Lund University

Teaching Assistant in Mathematics, Faculty of Engineering

Lund, Sweden

Aug. 2025 – Mar. 2026

- Led weekly exercise sessions where I taught calculus to a class of 30 engineering undergraduates

Axis Communications

Machine Learning Intern, Core Technologies Analytics

Lund, Sweden

June 2023 – May 2025

- Reduced bias in computer vision models using synthetic data
- Developed a pipeline for generating synthetic data in Unreal Engine
- Trained deep learning models for object detection and improved overall mAP, while reducing coverage bias by greatly improving performance on key object classes
- Implemented a novel noise reduction algorithm for pre-processing synthetic data in Python
- Streamlined the rendering pipeline in Unreal and refactored the codebase in C++
- Worked part-time during the terms and full-time during the summers

Hemkodat

Coding Instructor

Sweden

Sept. 2018 – Jan. 2023

- Travelled in Sweden during holidays and weekends to deliver courses in Python, Lua, and C# for adolescents

The Computer Engineering Guild at LTH

Vice President

Lund, Sweden

Jan. 2023 – Dec. 2023

- Led the board of a student guild with 1,000 members, chaired board meetings, and managed administration

SKILLS, PROJECTS & OTHER

Technical Skills: Python (PyTorch), Java, C++, C#, Scala, Haskell, SQL, CI/CD, Git, ROS2, Isaac Lab

Languages: Swedish native, English C2 (IELTS Academic 8.5/9.0), Spanish A2, learning French

Personal Projects: Created a VR simulation of a student dorm in Unreal, developed an algorithm for solving Ricochet robots in Java, built an NLP model with Keras, developed regression models for biological experiments

Open-Source: Working on an implementation of a more stable geodesic distance loss function in the PyRoMA framework for geometric deep learning with PyTorch

Awards: Won the national qualifiers and received a bronze medal at the European Union Science Olympiad (2019)